

Personal Technology: The Future Is You

Chapter 7



Chapter Topics

UNIT 7A: Personal Devices for Improving Productivity at School & Work

- 7.1 Convergence, Portability, & Personalization
- 7.2 Smartphones: More Than Talk
- 7.3 Tablets & E-Readers
- 7.4 Portable Media Players

UNIT 7B: personal Devices for Enriching Leisure & Life

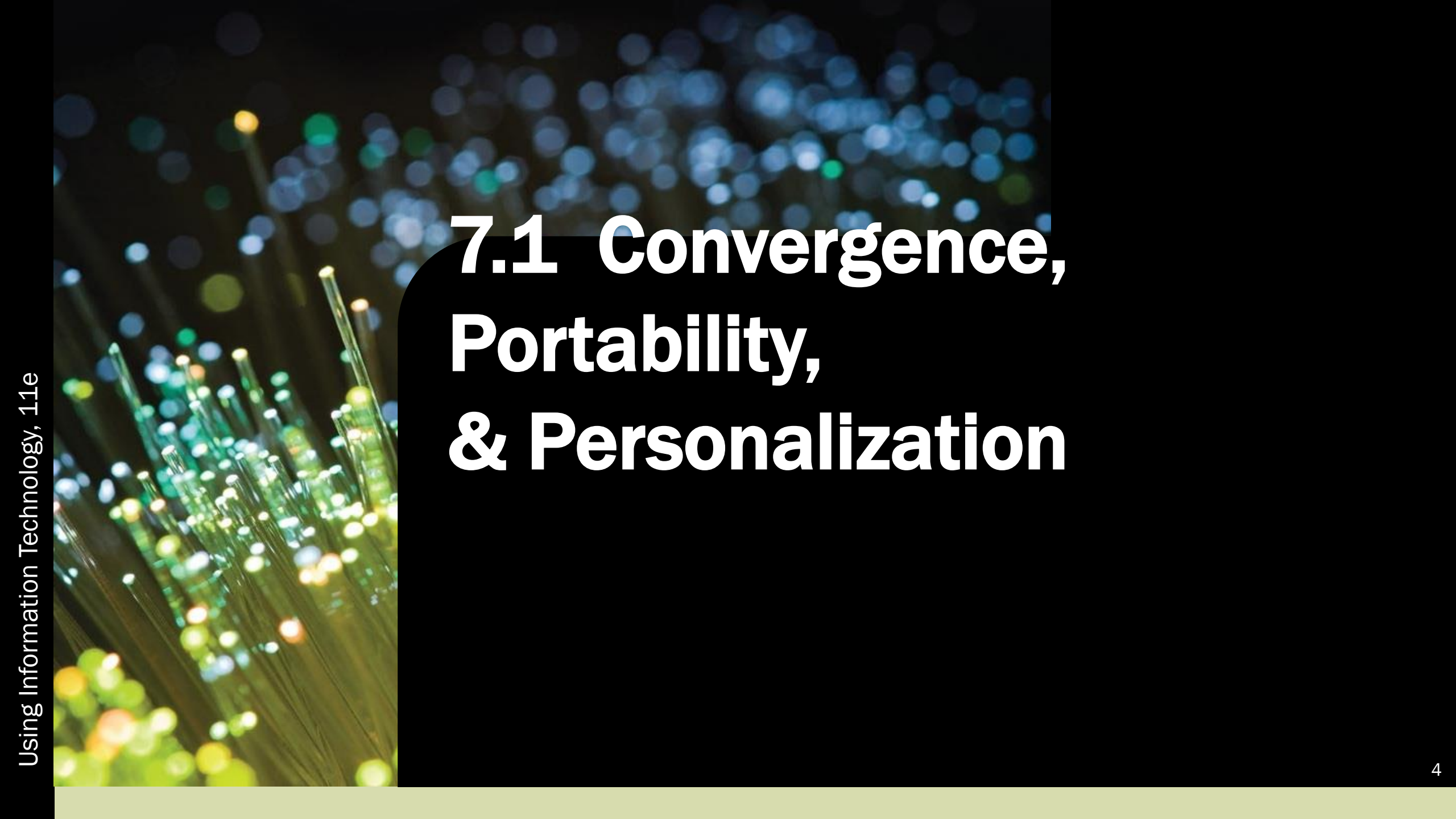
- 7.5 Digital Cameras: Changing Photography
- 7.6 High-Tech Radio: Satellite, Hybrid Digital, & Internet
- 7.7 Digital Television
- 7.8 Videogame Systems: The Ultimate Convergence Machine?



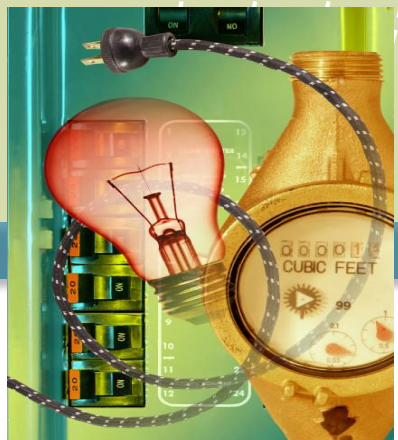


UNIT 7A: Personal Devices for Improving Productivity at School & Work

- The ongoing shift toward a more interactive and participatory web is exemplified by the **mashup**, a creative combination of content or elements from different sources, such as a web page that blends data from two or more sources to create new services or content.



7.1 Convergence, Portability, & Personalization



Three major trends in information technology continue to be convergence, portability, and personalization.

Convergence: Combining of several industries – computers, communications, consumer electronics, entertainment, and mass media – through various devices that exchange data in digital form

- *Pros:*
 - Increased convenience of devices; more affordable; more functions
- *Cons:*
 - Multiple features that compromise the primary feature—no single feature works optimally
 - Security risks are increasing



Portability

- *Pros*
 - Devices that enable phone, texting, and email access from anywhere, portable digital music, GPS, and convenient cheap digital photos that allow people to remain connected even while on the move
- *Cons*
 - Bombardment by texts and phone calls; intrusiveness; time wasters
 - Lack of face-to-face contact can lead to misinterpretations

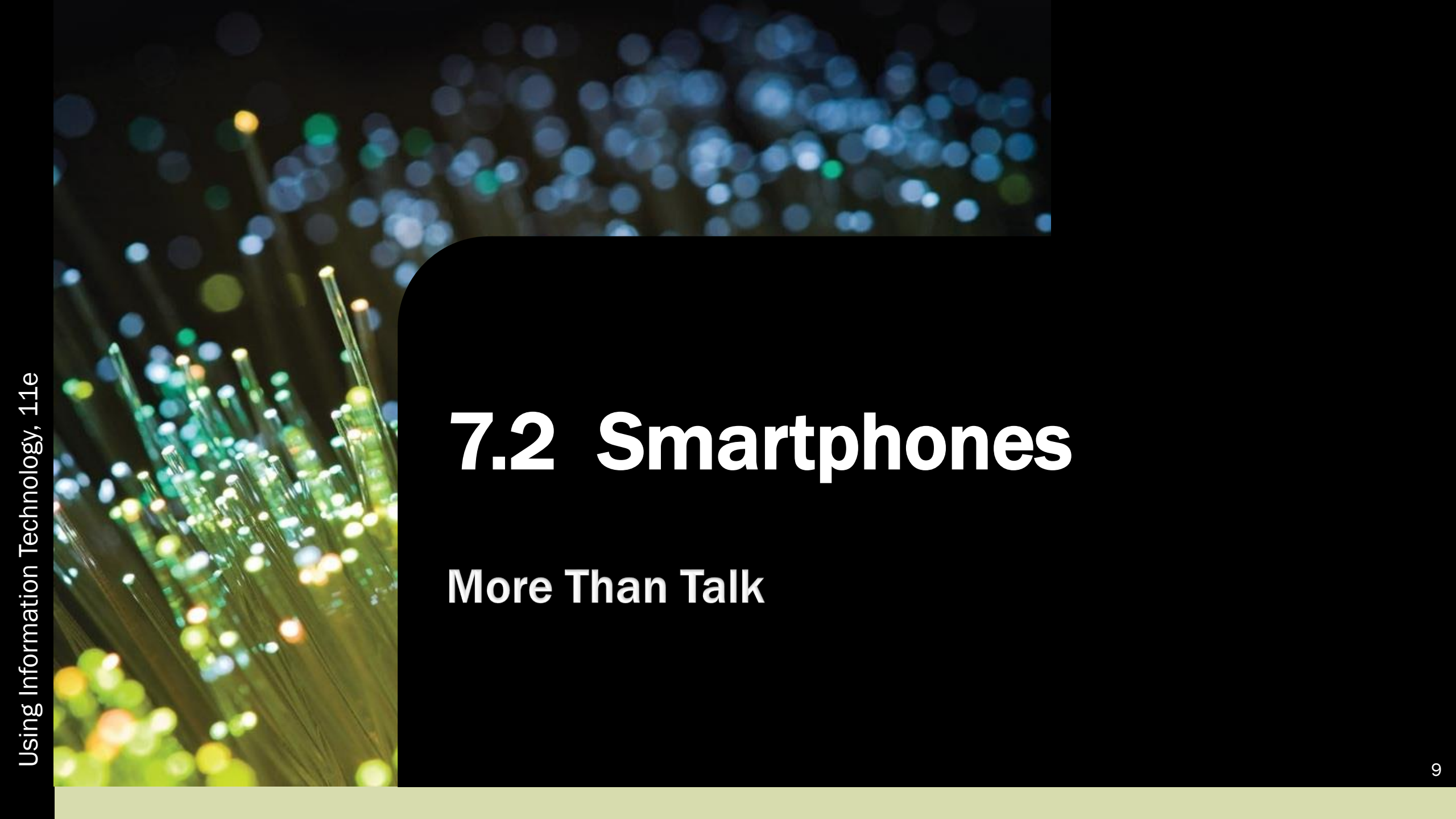


Personalization

- *Pros*
 - People can do many things, customized just for them. Downloaded hundreds or thousands of songs, so that they have their own personalized library of music; create lists of “favorites” or “bookmarks” so that they can readily access favorite website; access or contribute to blogs or personalized online diaries. In addition, PC software can be used to create all kinds of personal projects, ranging from artwork to finances to genealogy.
 - Get preselected news topics delivered to one’s electronic devices as needed.



- Personalization (*continued*)
 - Cons
 - Having many personalized devices leads to multitasking, which can lead to “absent presence” and nonfocus
 - Regret about choices
 - Inaction
 - Excessive expectations
 - Self-blame
 - “Paralysis” from too many choices
 - Filtering
 - Facts are facts: news should reflect the world, not us



7.2 Smartphones

More Than Talk



Smartphone = cellphone with microprocessor, memory, display screen, modem, apps, and Internet access

- Allows phone calls, email, web browsing, music availability, text messaging, videogames, digital TV viewing, search tools, GPS, personal information management, and so on
- *Storage*
 - Data is stored in flash memory card
 - Data does not disappear when phone is turned off





CELLPHONES: DESIGNED FOR CALLS & PERHAPS TEXTING

- Lower-priced than smartphones, cellphones are compact, have a straightforward keypad, let you receive and send text messages, and allow you to store frequently used phone numbers. Cellphones were originally designed for calls, and perhaps texting, but many have cameras, can access high-speed data networks, and support Bluetooth headsets for hands-free communication.

SMARTPHONES: DESIGNED WITH ADVANCED OPERATING SYSTEMS & TOUCH SCREENS

- Smartphones are more expensive than cellphones, have advanced operating systems (Apple's iOS, Google's Android, Windows Phone 8) that give them access to thousands of apps and have larger touch-screen displays and more advanced cameras than those on conventional phones, GPS navigation technology, Wi-Fi capabilities, and higher mobile wireless data speeds, which gives users better web browsing, search functions, and streaming media. They can also have a sophisticated organizer and allow you to do text messaging, handle multiple email accounts, review and edit documents, and create and edit spreadsheets. Many have media players that let you view videos and sort and listen to music tracks.



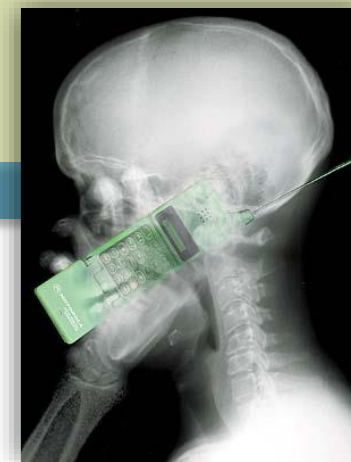
- **Cellphones are radios**
 - OS is stored in ROM; apps are downloaded
 - Display screens are 2 – 5 inches, color
 - Storage usually on flash memory cards
 - Have tactile or touch-screen keyboards
 - Microphone for voice commands
 - Receiver/Speaker to hear voice calls
 - Have headset connector
 - Are MP3/media players
 - Use Wi-Fi and/or Bluetooth
 - Have GPS and camera
 - Can show TV and video





- **Text messaging**, or *texting* or *SMS* (for short message service), is the sending of short messages, generally no more than 160 characters in length (including spaces), to a smartphone or other handheld device.





- Societal Effects of Cellphones

- Positive (among others)
 - Parents can more easily monitor their children
 - Police dispatchers can help people who are lost
 - Information and amusements are readily available
 - Get road assistance
 - Information can get out quickly during emergencies
- Negative (among others)
 - People are less polite, courteous, and respectful using cellphones
 - Cellphones are answered in theaters and restaurants
 - Cellphone users engage in loud conversations in public places
 - Cellphone use while driving is dangerous



7.3 Tablets & E-Readers

A **tablet** is a general-purpose computer contained in a single panel; it is a combination of smartphone and laptop computer with wireless connections, a 7- to 12-inch multitouch screen, and a virtual screen.

- *Platforms (OS):* iPad, Android, Windows RT





E-readers are better than tablets for reading e-books in terms of price, screen quality, simplicity and portability.

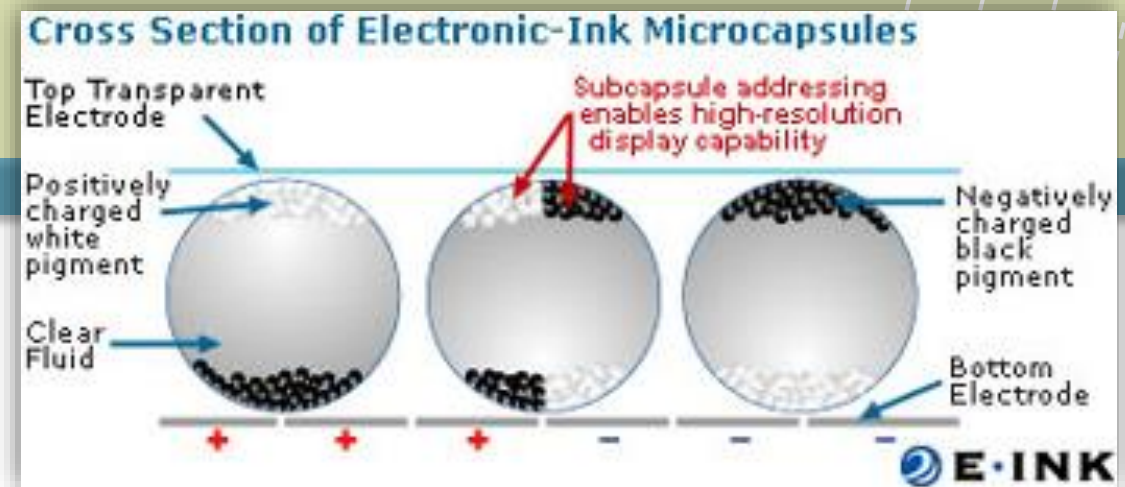
- **E-Book**, or electronic book: electronic text, the digital-media equivalent of a conventional printed book
- **E-Book Reader**: electronic device that can download books (wired and wirelessly)
- Examples: Kindle (Amazon); Nook (Barnes & Noble)



- Some Benefits of E-Readers

- One e-book reader can store hundreds or thousands of books.
- Easy to download books by wireless access; books are less expensive.
- Type size and face can be adjusted.
- Usually can be read in low light.
- Automatically opens to the page where you left off.
- Text can be searched and cross-referenced.
- Dictionary automatically available.





How an E-Reader Works

- E-readers use E Ink, which is composed of millions of tiny particles that display text.
- Different e-books use different e-software formats.
- E-books are downloaded by wireless access (3G or Wi-Fi).



Some drawbacks of e-readers:

- Photos, charts, diagrams, foreign characters, and tables not as good as in print version—or are left out of the e-book
- Battery needs to be recharged
- Reader doesn't own the downloaded books
- Are not always designed for college textbook use
- Can be hacked
- Can't lend your books
- They can malfunction



7.4 Portable Media Players



Portable media players include music players, media players, and some mobile phones.

- **Portable media players (PMPs)** are portable devices that play digital audio, video, and/or image files.
- **MP3** is a format that allows audio files to be compressed so they are small enough to be sent over the Internet and stored as digital files.





PMP Technology Considerations: What's Useful to Know?

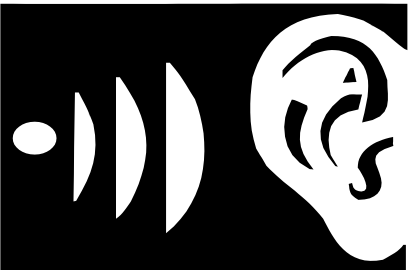
- Storage capacity (both hard drive and flash memory)
- Sampling rate
- Transferring files
- Battery life
- Display screens
- Other features such as:
 - FM radio reception
 - Music recording using microphone
- Getting music and video files
- Using PMP in a car





Societal Effects

- PMPs offer convenience and portability to music listeners.
- May also cause people to isolate themselves.
- MP3 players are used almost everywhere by many people, but they do have the ability to damage hearing, so users should be careful with the volume controls. Over 85 decibels can cause hearing loss! (85 decibels is as loud as a vacuum cleaner or a crowded restaurant .)
- Over-the-ear headphones are safer than earbuds.





UNIT 7B: Personal Devices for Enriching Leisure & Life





7.5 Digital Cameras

Changing Photography



A **digital camera** takes video and photographs and digitally converts the analog data by recording images via an electronic image sensor (they do not require film).

- **Point-and-shoot** digital camera
 - Automatically adjusts settings such as exposure and focus
 - Easy to use, but manual controls can allow you to tweak the settings to get better photos
- **Single-lens reflex (SLR)** digital camera
 - Uses a reflecting mirror to reflect the incoming light so the viewfinder shows what the lens is framing
 - Brighter and crisper photos
 - Used by professionals





Resolution: Megapixels and Sensors

- **Megapixels** = millions of picture elements (electronic dots making up an image); the more megapixels, the higher the resolution
- 10 – 16 megapixels common for point-and-shoot cameras; 10 – 24 for SLR
- The larger the **sensor** chip, the sharper the images

Lenses

- Digital zoom
 - Means the image is cropped in the camera
 - Lowers the resolution and so can produce a grainy photo
- Optical zoom
 - Enlarges the subject without you needing to move closer
 - Lens extends to focus on distant objects; clearer image





Storage

- Use flash memory cards inside camera; cards are reusable
- 64 MB holds only a handful of images; 512 MB – 1 GB is better

Viewing images

- Optical viewfinders let you see the image to be photographed before you snap the picture
- LCD screens let you review the photos you have taken



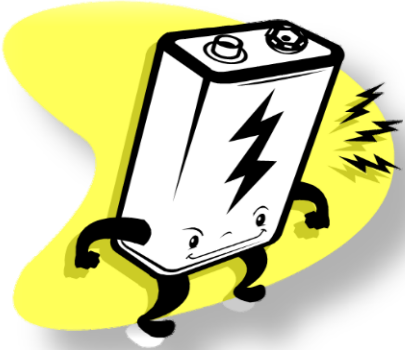
- **Start-up time**

- Digital cameras require time to start up
- Look for one that has a short start-up time
- Also, the shutter can lag, delaying the time between when you press the button and the shutter clicks; look for a camera that allows “burst” or “continuous “ mode
- Continuous mode allows you to take a certain number of shots without pausing or reclicking a button



- **Battery life**

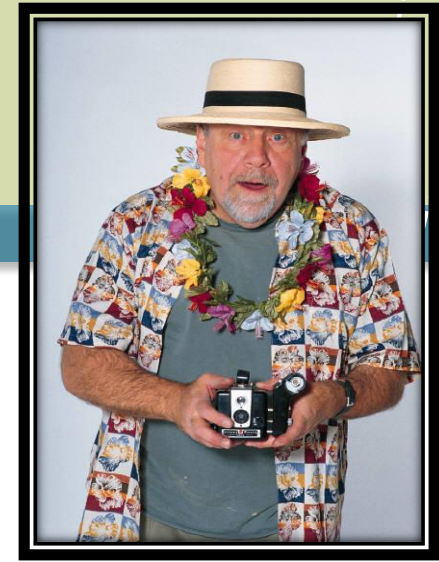
- The camera requires a battery to function
- Some rechargeable batteries are available with many models
- Some recharge in the camera, while others require a separate charging stand
- Get a battery that can last a whole day
- Carry a spare battery





Principle methods for transferring images

- Use a direct connection between your camera and your computer (with a USB cable and software)
- Wireless connection
- Insert the memory card into your computer or card reader
- Put your camera into a cradle attached to the PC
- Use an online photo service
- Use a photo printer with a built-in card slot
- Use a photo-printing kiosk
- Use a photo lab
- Bring along your own card reader and CDs and use others' computers



Societal Effects of Digital Cameras

- People are taking their cameras everywhere
- People take far more pictures than they used to
- Photography is becoming more casual
- People are touching up their photos, making them look better
- Camera use can be intrusive and even illegal (voyeurism)

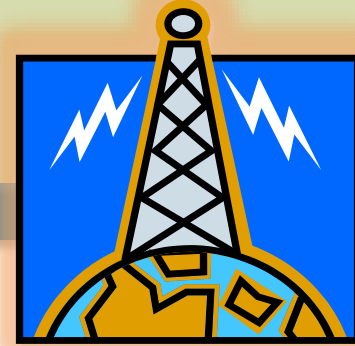
7.6 High-Tech Radio

Satellite, Hybrid Digital, & Internet

Satellite radio

- Digital radio signals are sent from satellites in orbit around the earth to subscribers that have special radios, often in cars.
- CD-quality sound is better than normal radio.
- More channels than regular radio.
- U.S. provider is Sirius XM satellite radio.
- Commercial-free.





HD (Hybrid Digital) Radio

- Provides CD-quality sound on the FM dial
- Allow broadcasters to squeeze one analog and two digital stations on the same frequency
- Broadcasts are free—no subscription charges
- Broadcasters are hoping HD radio can introduce more local or innovative programming

Internet Radio

- Continuous streaming of audio over the Internet. Internet users can listen to radio on their computers and various handheld devices, such as a smartphone (Pandora, iHeartRadio, Slacker)



Podcasting

- Involves the recording of Internet radio or similar Internet programs
- Requires no studio or broadcast tower and is not regulated by the FCC (Federal Communications Commission)
- Allows amateur deejays and hobbyists to create their own radio shows and offer them on the Internet
- Podcasting-receiving software, called an **aggregator**, is necessary



7.7 Digital Television



Digital Television:

- *Interactive TV*
 - Lets you interact with the show you're watching
- *Internet TV*
 - Television distributed via the Internet, viewable on computers and mobile devices
- *Internet-Ready TV*
 - TVs with broadband modems allow viewers to watch TV shows as well as go online to browse, get news, stream movies, view photos, etc.



3 Kinds of Digital TV Broadcasting

- **Digital television (DTV):** uses a digital signal (not analog)
 - Is clearer and less prone to interference than analog
 - FCC has mandated that all TV stations be capable of digital broadcasting
 - People with analog TVs use a converter box to deal with digital broadcast signals
- **High-definition television (HDTV)**
 - Works with digital broadcasting signals
 - Has broader screen and higher resolution than analog TV
 - Uses a lot of bandwidth
- **Standard-definition television (SDTV)**
 - Uses lower resolution than HDTV and so can transmit more information within the HDTV bandwidth
 - Allows multicasting – up to 5 SDTV programs at the same time



- **Societal Effects of Digital TV**

- *Time shifting*: Changing when you watch TV
- *Space shifting*: Changing where you watch TV
- *Content shifting*: Changing the nature of TV programs
- **Video on demand** (VOD or VoD) consists of a wide set of technologies that enable viewers to select videos or TV programs from a central server to watch when they want, rather than when TV programmers offer them



7.8 Videogame Systems

The Ultimate Convergence Machine?

Videogame consoles, or gameboxes, are specialized computers powered by operating systems and CPUs different from those in desktop computers.

- These may be the “ultimate convergence machine.”
- People buy them to play games, but they do a lot more: play music, share photos, watch movies
 - Microsoft Xbox
 - Sony PlayStation 4
 - Nintendo’s Wii U

